

CS516: Parallelization of Programs

Optimizing Data Transfers

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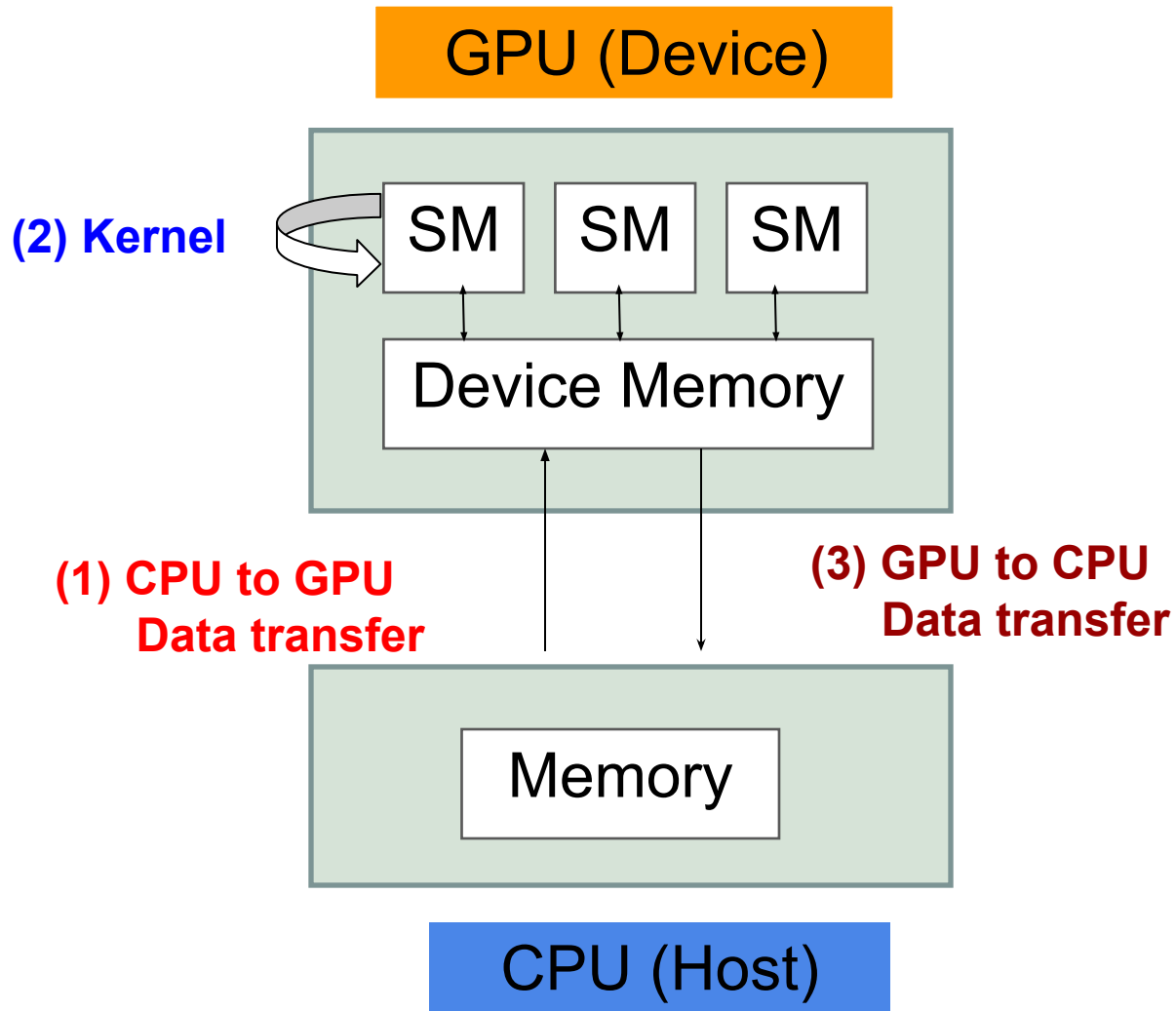
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Outline

- Optimizing Memory Transfers
- Unified Memory

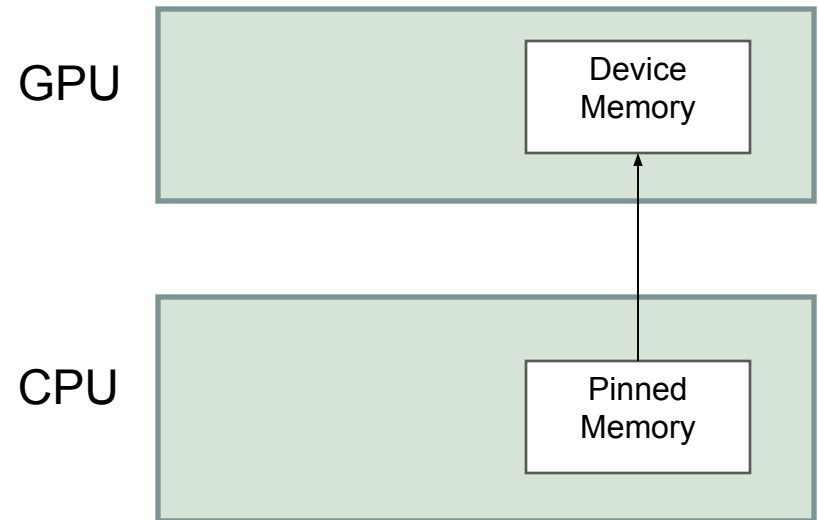
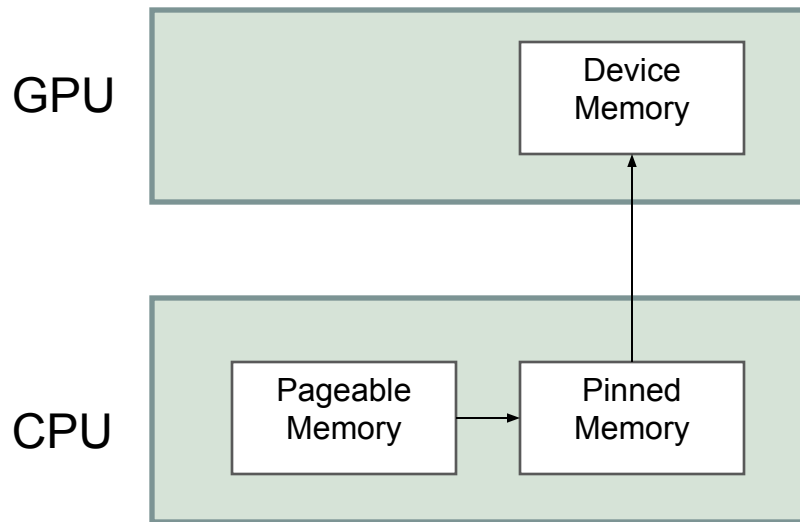
CUDA Program Flow



Optimizing Memory Transfer – Pinned Memory

```
int main() {  
    h_a = malloc(..) //host pageable  
    d_a = cudaMalloc(..) //device  
    /* Initialize h_a */  
    cudaMemcpy(d_a, h_a, HostToDevice)  
    /* Process d_a */  
}
```

```
int main() {  
    h_a=cudaMallocHost(..) //host pinned  
    d_a = cudaMalloc(..) //device  
    /* Initialize h_a */  
    cudaMemcpy(d_a, h_a, HostToDevice)  
    /* Process d_a */  
}
```



Overlapping Computation and Communication

```
cudaMemcpy(d_a, a, numBytes, cudaMemcpyHostToDevice);  
increment<<<1,N>>>(d_a)  
cudaMemcpy(a, d_a, numBytes, cudaMemcpyDeviceToHost);
```

Blocking

Asynchronous

Blocking

```
cudaMemcpy(d_a, a, numBytes, cudaMemcpyHostToDevice);  
increment<<<1,N>>>(d_a)  
Compute()  
cudaMemcpy(a, d_a, numBytes, cudaMemcpyDeviceToHost);
```